Weak Acids and Ka

Ka, pKa and pH of weak acids and bases

Name _____ Class _____ Date _____

DATA (25 °C)

K_a acetic acid = 1.8×10^{-5} K_a HF = 6.8×10^{-4} K_a HCN = 4.9×10^{-10}

K_b NH3 = 1.8×10^{-5} $K_w = 1.0 \times 10^{-14}$

Use the shortcut [H3O+] ≈ √(Ka × C) and check that less than 5% ionizes. Give pH to two decimal places.

A. Understanding

1. Write the K_a expression for a weak acid HA, and explain what a large K_a tells you.

2. Rank HF, acetic acid and HCN from strongest to weakest acid, and calculate the pKa of HF.

B. pH of weak acids

3. Calculate the pH and percent ionization of 0.20 M acetic acid.

4. Calculate the pH of 0.25 M hydrocyanic acid, HCN.

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C. Ka from a measured pH

5. A 0.050 M solution of a weak acid has pH 3.20. Calculate K_a and the percent ionization.

D. Weak bases

6. Calculate the pH of 0.20 M ammonia.

7. Calculate K_b for the acetate ion, the conjugate base of acetic acid.

E. Challenge

8. Explain why 0.20 M acetic acid and 0.20 M hydrochloric acid have very different pH values, even though both are 0.20 M acids.

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